A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

Impulse & 2D Collisions

At AS you saw that momentum is conserved when things collide in a straight line. This topic finishes the story. First we look at what a force does to momentum over time, which is called **impulse** and explains crumple zones, airbags and why you bend your knees when you land. Then we take collisions off the line and onto a table top, where balls fly off at angles, and learn how to tell whether any kinetic energy was lost on the way.

SPEC 81 Impulse and change of momentum

WHAT THIS SPEC POINT ASKS

Use impulse = FΔt = Δp, which is Newton's second law written in terms of momentum.

Momentum

FORMULA SHEET

$$p = mv$$

p: momentum / kg m s^{-1} m: mass / kg v: velocity / m s^{-1}

Force is the rate of change of momentum

- Momentum p = mv (unit kg m s^{-1} or N s). It is a **vector**, so its direction matters.
- Newton's second law in its general form: the **resultant force** equals the rate of change of momentum.

Impulse

FORMULA SHEET

$$F\Delta t = \Delta p$$

F: resultant force / N Δt: time the force acts / s Δp: change in momentum / kg m s^{-1} or N s

From F = ma to F = Δp/Δt

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Start with Newton's second law for a constant resultant force F acting on a mass m:

$$F = ma$$

2. The force is constant, so the acceleration is constant. It is the change in velocity divided by the time taken, where u is the velocity before and v the velocity after:

$$F = m\,\frac{v - u}{\Delta t}$$

3. Multiply out the bracket. The mass is constant, so it can go inside:

$$F = \frac{mv - mu}{\Delta t}$$

mv is the final momentum and mu the initial momentum, so the top line is the change in momentum Δp.

4. So the resultant force equals the rate of change of momentum:

$$\boxed{F = \frac{\Delta p}{\Delta t}}$$

We assumed the mass stays the same. $F = \Delta p / \Delta t$ still works when it doesn't, which is why Newton's own version is the more general one.

- F = average **resultant** force (N), Δt = time the force acts (s), m = mass (kg), u and v = initial and final velocity (m s^{-1}).
- **Impulse** = FΔt. Its unit is the **N s**, which is the same as kg m s^{-1}. Impulse points the same way as the force.
- If the mass is constant, FΔt = m(v − u) becomes F = ma, the form you met at AS.

Force–time graphs

- If the force changes during the impact, the **area under a force–time graph** is the impulse, so it equals the change in momentum.
- For a non-constant force, FΔt uses the **average** force. The peak force is bigger.

Why longer contact times mean smaller forces

For a fixed change in momentum, FΔt is fixed, so a **longer** collision time gives a **smaller** average force. Crumple zones, airbags, seat belts, helmets, crash mats and bending your knees on landing all work this way: they **deform**, which **increases the time** of the collision, so for the same Δp the force is **reduced**.

WORKED EXAMPLE

ball rebounding from a wall

A 0.058 kg tennis ball hits a wall at 25 m s^{-1} and rebounds at 18 m s^{-1}. Contact lasts 4.0 ms. Find the average force on the ball.

Take **away from the wall as positive**: u = −25 m s^{-1}, v = +18 m s^{-1}.

Δp = m(v − u) = 0.058 × (18 − (−25)) = 0.058 × 43 = 2.49 N s

F = Δp ÷ Δt = 2.49 ÷ 4.0 × 10^{-3} = **620 N** (2 s.f.), directed away from the wall.

If you forget the sign change you get 0.058 × 7 = 0.41 N s, which is wrong. A rebound always **adds** the speeds.

WORKED EXAMPLE

A tennis ball of mass 0.058 kg hits a wall at 25 m s^{-1} and rebounds along the same line at 20 m s^{-1}. It is in contact with the wall for 4.0 ms. Calculate (a) the impulse on the ball and (b) the average force the wall exerts on it.

Show solution

(a) Choose a positive direction first: **towards the wall is positive**. Then $u = +25\ \text{m s}^{-1}$ and $v = -20\ \text{m s}^{-1}$.

$$\Delta p = m(v - u) = 0.058 \times (-20 - 25) = 0.058 \times (-45) = -2.61\ \text{N s}$$

The impulse is $2.6\ \text{N s}$ directed away from the wall.

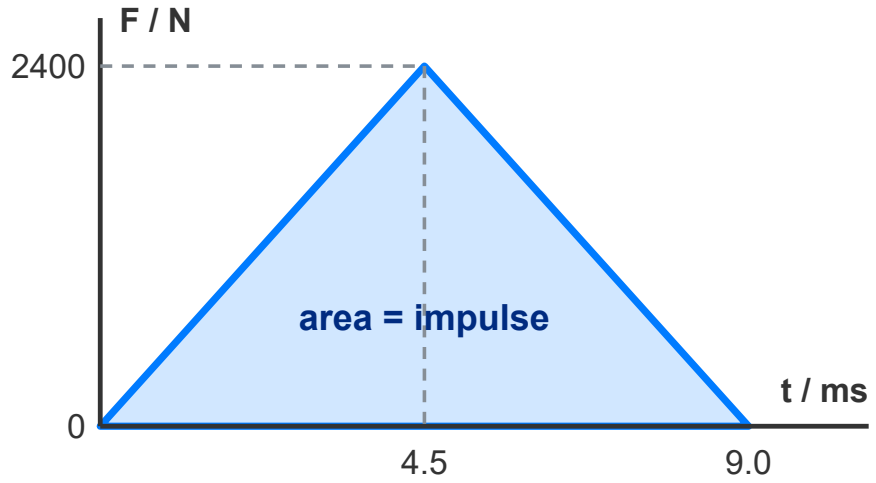
(b)

$$F = \frac{\Delta p}{\Delta t} = \frac{-2.61}{4.0 \times 10^{-3}} = -653\ \text{N}$$

So the average force is about $650\ \text{N}$ away from the wall. That is over a thousand times the ball's weight, which is why the ball squashes visibly.

WORKED EXAMPLE

A football of mass 0.45 kg is at rest. The graph shows how the force from a boot varies during the kick. Find the speed of the ball as it leaves the boot, and the average force during the kick.



Force on the ball during the kick.

Show solution

The impulse is the area of the triangle. Convert the time to seconds first:

$$F\Delta t = \tfrac{1}{2} \times 9.0 \times 10^{-3} \times 2400 = 10.8 \text{ N s}$$

The ball starts at rest, so all of this is its final momentum:

$$v = \frac{\Delta p}{m} = \frac{10.8}{0.45} = 24 \text{ m s}^{-1}$$

The average force is the impulse spread over the whole contact time:

$$F_{\text{avg}} = \frac{10.8}{9.0 \times 10^{-3}} = 1200 \text{ N}$$

For a triangle the average force is exactly half the peak, which is a nice check.

MARK-SCHEME WORDING

- "Use of FΔt = Δp" and "use of Δp = mΔv" are separate method marks.
- Cushioning explanations need three linked steps: "material **deforms**" → "**time** of collision **increases**" → "since FΔt = Δp (and Δp is the same), the **force is reduced**". Name the change in momentum.
- Newton's third law pairs: "equal and opposite forces on different objects". For a hovering helicopter, the rotor pushes air down, the air pushes the rotor up; "resultant force is zero, so no acceleration".

Often asked as: force from a momentum change, the area under an F–t graph (MCQ), why a rubber strip or helmet reduces the force, a ball bouncing back (Δp = 2mv for equal speeds), unit of impulse (N s).

COMMON MISTAKE

Saying the force is smaller "because the change in momentum is smaller". Δp is the **same**; it is the **time** that is longer.

Explaining with "energy absorbed" or "friction" but never mentioning time.

Forgetting the sign change on a rebound. Write down your positive direction first.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Use the resultant force.** If the weight also acts during the impulse (a jump, a landing), FΔt = Δp applies to the **resultant** force: e.g. contact force − weight (WPH14 Jun 2022).
- **Fluid streams.** For air or water hitting a surface, mass per second = ρAv. The force needed to stop it is (mass per second) × v. Example: a water jet of area 2.0×10^{-4} m² at 15 m s^{-1} (ρ = 1000 kg m^{-3}) hits a wall and stops. Mass per second = $1000 \times 2.0 \times 10^{-4} \times 15 = 3.0$ kg s^{-1}, so F = 3.0 × 15 = **45 N** (WPH14 Oct 2024).
- **Compare with a threshold.** Work out an impulse or force and compare it with a stated value, then say whether a claim holds (WPH14 Jan 2026).

SPEC 82 Core Practical 9: force and change of momentum

WHAT THIS SPEC POINT ASKS

Carry out an experiment to investigate how the force on an object is related to its change of momentum.

Principle

A constant resultant force F acting for a time Δt should give a change in momentum Δp = FΔt. If you measure Δp for different times (or different forces), a graph of Δp against Δt is a **straight line through the origin with gradient F**.

CORE PRACTICAL

Core practical 9: method

Apparatus: dynamics trolley with a card of known length on top, runway, two light gates connected to a data logger (or timer), pulley, string, slotted masses on a hanger, electronic balance, metre rule.

1. Measure the mass of the trolley M and the hanging mass m with the balance.
2. **Compensate for friction:** tilt the runway slightly until the trolley, given a small push, rolls at constant speed (equal times at both light gates).
3. Attach the string over the pulley to the hanger. The accelerating force is the weight of the hanger, F = mg.
4. Release the trolley from rest. The light gates record the time for the card to pass each gate (speed = card length ÷ time) and the time Δt between the gates.
5. Repeat for several gate separations (or several hanging masses), three times each, and find means.
6. Calculate $\Delta p = (M + m)(v_2 - v_1)$ for each run and $F\Delta t = mg\Delta t$.

Variables

Variable	
Independent	time between the gates Δt (or the accelerating force mg)
Dependent	change in momentum Δp, from the two speeds
Control	total mass (M + m), card length, friction compensation, release point

Processing the results

- The force mg accelerates **both** the trolley and the hanging mass, so use the **total** mass:

Impulse on the trolley and hanger

$$mg\,\Delta t = (M + m)(v_2 - v_1)$$

m: hanging mass / kg M: trolley mass / kg Δt: time between the gates / s

v_1, v_2: speeds at the two gates / m s^{-1}

- Plot Δp (y-axis) against Δt (x-axis). Gradient = force. Compare it with mg. Or plot Δp against FΔt: gradient should be 1.

WORKED EXAMPLE

checking FΔt = Δp

Trolley M = 0.800 kg, hanging mass m = 0.050 kg, so total mass 0.850 kg and F = 0.050 × 9.81 = 0.491 N.

Speeds at the gates: v_1 = 0.500 m s^{-1}, v_2 = 1.000 m s^{-1}. Time between gates Δt = 0.87 s.

Δp = 0.850 × (1.000 − 0.500) = **0.43 N s** (2 s.f.)

FΔt = 0.491 × 0.87 = **0.43 N s** (2 s.f.)

The two agree, so the change in momentum equals the impulse.

Errors and improvements

- **Friction** not fully compensated: the resultant force is less than mg, so Δp is smaller than FΔt (a systematic error).
- Using only the trolley's mass instead of M + m gives a gradient that is too small.
- Card not perpendicular to the beam, or a card that is too short, gives poor speed readings. Measure the card length with vernier calipers.
- Repeat and average; discard runs where the trolley swerves or touches a gate.
- **Alternative method:** let a trolley hit a **force sensor** on a data logger. The area under the force–time graph (impulse) is compared with m(v − u) from light gates before and after the collision.

EXAM TIP

This practical has not been set as a "describe the method" question so far. It turns up as data: light-gate speeds turned into momentum, a collision time, or an area under an F–t graph. Make sure you can turn a card length and a gate time into a speed, then into momentum.

COMMON MISTAKE

Writing the equation as m t = Mg(v_2 − v_1) (seen in some revision sources). The force is the **hanging weight** mg; the mass being accelerated is (**M + m**). Get the units of each side to match: N s on both.

SPEC 83 Conservation of linear momentum in two dimensions

WHAT THIS SPEC POINT ASKS

Apply conservation of linear momentum to collisions and explosions in two dimensions.

The principle

- **Total momentum before = total momentum after**, provided **no resultant external force** acts (a closed system).
- Momentum is a vector, so in 2-D it is conserved **separately in each of two perpendicular directions**.
- Weight and the normal contact force still act on pucks on a table, but they cancel. Friction is the force you have to assume is negligible.

Method for a 2-D collision

1. Choose axes: x along the original direction of motion, y perpendicular to it.
2. Resolve every momentum after the collision: p cos θ along x and p sin θ along y, with θ measured from the x-axis. Check which line each angle is measured from.
3. **y-direction:** total y-momentum before is usually **zero**, so the y-components after must cancel. This often gives the first unknown.
4. **x-direction:** total x-momentum before = sum of x-components after.
5. Combine components with **Pythagoras** (magnitude) and **tan θ = p_y / p_x** (direction).

You can also draw a **vector diagram**: momentum vectors drawn to scale, tip to tail. If momentum is conserved, the vectors after add to the vector before, so the diagram **closes into a triangle**.

WORKED EXAMPLE

puck collision

Puck A (0.20 kg) moving at 3.0 m s^{-1} hits stationary puck B (0.30 kg). Afterwards A moves at 1.5 m s^{-1} at 50° to its original line. Find B's velocity.

Momentum before: x: 0.20 × 3.0 = 0.60 N s; y: 0.

A after: x: 0.20 × 1.5 cos 50° = 0.193 N s; y: 0.20 × 1.5 sin 50° = 0.230 N s.

x: 0.60 = 0.193 + p_{Bx} → p_{Bx} = 0.407 N s

y: 0 = 0.230 − p_{By} → p_{By} = 0.230 N s (on the other side of the line)

$p_B = \sqrt{(0.407^2 + 0.230^2)}$ = 0.468 N s, so v_B = 0.468 ÷ 0.30 = **1.6 m s^{-1}**

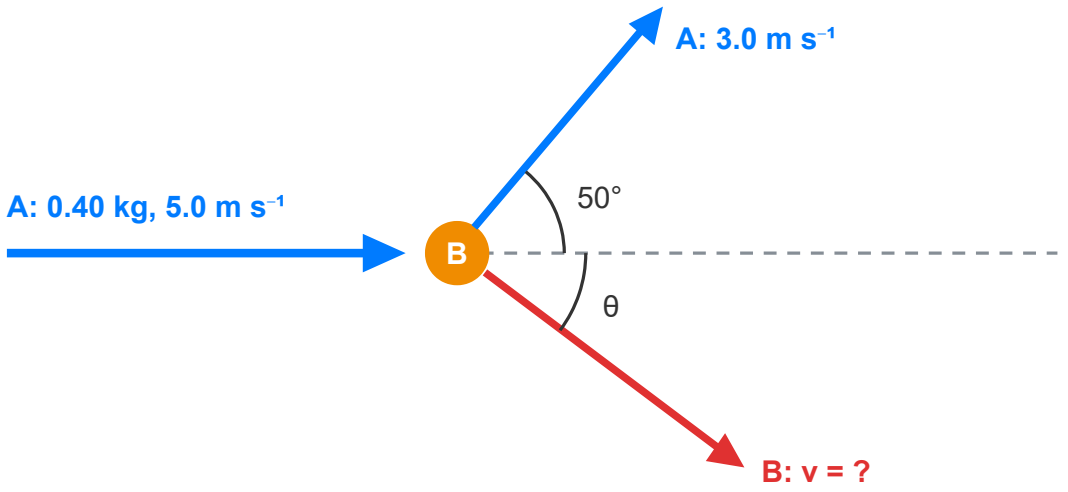
Angle: tan θ = 0.230 ÷ 0.407 → θ = **29°** to the original line, on the opposite side from A.

Explosions and recoil

- If everything starts at rest, total momentum is zero, so the fragments' momenta add to zero. Two fragments move off in **exactly opposite** directions with **equal-sized momenta**: $m_1v_1 = m_2v_2$.
- In a radioactive decay with three products (or a recoiling nucleus and two particles), the three momenta form a **closed triangle**.

WORKED EXAMPLE

Puck A (mass 0.40 kg) slides at 5.0 m s⁻¹ and hits puck B (mass 0.60 kg), which is at rest. Afterwards A moves at 3.0 m s⁻¹ at 50° to its original direction. Find (a) the speed and direction of B and (b) whether the collision is elastic.



Take x along A's original direction and y upwards.

Show solution

x-direction. Before: $0.40 \times 5.0 = 2.00\ \text{kg m s}^{-1}$. After, A contributes $0.40 \times 3.0\cos 50^\circ = 0.771\ \text{kg m s}^{-1}$, so

$$0.60\,v_{Bx} = 2.00 - 0.771 = 1.229 \quad \Rightarrow \quad v_{Bx} = 2.05\ \text{m s}^{-1}$$

y-direction. Before: zero. After, A contributes $+0.40 \times 3.0\sin 50^\circ = +0.919\ \text{kg m s}^{-1}$, so B must have -0.919:

$$0.60\,v_{By} = -0.919 \quad \Rightarrow \quad v_{By} = -1.53\ \text{m s}^{-1}$$

Combine the components with Pythagoras and tan:

$$v_B = \sqrt{2.05^2 + 1.53^2} = 2.56\ \text{m s}^{-1}, \qquad \theta = \tan^{-1}\left(\frac{1.53}{2.05}\right) = 36.8^\circ$$

So B moves at $2.6\ \text{m s}^{-1}$ at 37° below A's original direction.

(b) Kinetic energy is a scalar, so just add the values for each object, using the full speeds:

$$E_{\text{before}} = \tfrac{1}{2}(0.40)(5.0)^2 = 5.0\ \text{J}$$

$$E_{\text{after}} = \tfrac{1}{2}(0.40)(3.0)^2 + \tfrac{1}{2}(0.60)(2.56)^2 = 1.80 + 1.96 = 3.8 \text{ J}$$

Kinetic energy has fallen by about 1.2 J (a quarter of it), so the collision is **inelastic**.

MARK-SCHEME WORDING

- Condition: "total momentum before = total momentum after, **if no (resultant) external force** acts" (or "in a closed system").
- For pucks on ice or air tables: "no / negligible friction", so "**no resultant external force**". "Reduces friction" alone is not enough.
- Marks: "use of p = mv", "use of trigonometry to find a component", "conservation in one direction", then "in both directions". A final comparison mark wants "momentum before = momentum after (in both directions)".
- Vector diagram: "vectors drawn to scale", "end to end", "with arrows", "a **closed triangle**, so momentum is conserved".

Often asked as: pucks, curling stones, cars at a junction, ships, alpha particles scattering off nuclei, decay with a recoiling nucleus.

COMMON MISTAKE

Conserving **velocity** instead of momentum, or calculating kinetic energy where momentum was wanted (almost a fifth of one cohort did this).

Swapping sin and cos. If θ is measured from the x-axis, the x-component uses cos θ.

Stopping after the x-equation. A 2-D question needs both directions to finish.

Cancelling the masses without showing it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Equal masses, elastic collision, one at rest:** the two bodies move off **at 90° to each other**. Questions ask you to test this from measured angles (WPH14 Jun 2024).
- **Alpha tracks with a recoil:** a short gap at the start of a track shows the nucleus recoiled between two decays (WPH14 Oct 2020).

SPEC 84 Core Practical 10: analysing collisions with ICT

WHAT THIS SPEC POINT ASKS

Use ICT (video and analysis software) to analyse collisions between small spheres, such as ball bearings on a table top.

The idea

Film a collision between two spheres from above, then use video-analysis software (e.g. Tracker) to find each sphere's velocity before and after, frame by frame. Then test conservation of momentum in two directions, and test whether kinetic energy is conserved.

CORE PRACTICAL

Core practical 10: method

1. Measure the mass of each sphere on an electronic balance.
2. Place a metre rule (or a scale of known length) flat in the field of view, at the same level as the spheres, for calibration.
3. Mount the camera (or phone) **directly above** the table, looking straight down, so the view is perpendicular to the surface (no parallax).
4. Roll one sphere (e.g. down a short ramp for a repeatable speed) into a stationary sphere so that they hit off-centre and move off at angles.
5. Load the video into the software. Set the scale from the rule and the time between frames from the frame rate. Mark each sphere's position frame by frame.
6. Read off the velocity components (or speed and angle) before and after. Repeat for different impact angles.

Low-tech alternative: put carbon paper under paper on the table so the spheres leave tracks; measure angles with a protractor. Speeds can be found from distance travelled between known times.

Analysis

- Calculate p = mv for each sphere and resolve into components along and perpendicular to the original direction (as in 83).
- Check: x-momentum before ≈ x-momentum after; y-components after ≈ cancel.
- Check kinetic energy before and after to decide elastic or inelastic (see 85).
- Compare each measured value with the calculated one and say if they agree **within experimental uncertainty**. A difference of about 1° in an angle is "consistent".

Errors

Source	Type	Fix
Camera not directly above, or scale not at the spheres' height	systematic (parallax / scale)	mount camera vertically; scale in the same plane
Frame rate too low; blurred images	random	higher frame rate, faster shutter, good lighting
Friction and rolling (some energy goes into rotation)	systematic	smooth level surface; hard spheres
Spheres not hitting cleanly	random	guide ramp; repeat runs

MARK-SCHEME WORDING

- Measuring angles: "film from directly **above**", "measure angles on playback with a **protractor**" (or motion-tracking software), or "carbon paper to record tracks, then protractor".
- Comparison: "compare the student's measured value with the calculated value", then "**conclusion**: consistent / not consistent".

Often asked as: how to measure the angles; use given angles and speeds to test conservation of momentum; is the collision elastic?

EXAM TIP

State the conclusion explicitly and give both numbers with units. "41.5° compared with a measured 43°, which agrees within the uncertainty of a protractor reading, so momentum is conserved."

SPEC 85 Elastic and inelastic collisions

WHAT THIS SPEC POINT ASKS

Decide whether a collision is elastic or inelastic.

Momentum always, kinetic energy sometimes

- In **every** collision (with no external force), **momentum is conserved** and **total energy is conserved**.
- **Elastic collision:** total **kinetic energy** is also conserved (Ek before = Ek after).
- **Inelastic collision:** some kinetic energy is transferred to other stores (internal energy/heating, sound, deforming the objects), so Ek after < Ek before.

- When objects **stick together** the collision is inelastic, and the most Ek is lost that momentum conservation allows.
- In an **explosion**, Ek increases (energy comes from a chemical or nuclear store), but momentum is still conserved.

How to test

1. Calculate Ek = ½mv² (or p²/2m) for **every** body before the collision and add them.
2. Do the same after.
3. **Compare** the two totals, with units, and state the conclusion.

WORKED EXAMPLE

was the puck collision elastic?

Using the puck collision in 83 (A: 0.20 kg, 3.0 m s^{-1} → 1.5 m s^{-1}; B: 0.30 kg, 0 → 1.56 m s^{-1}).

Ek before = ½ × 0.20 × 3.0^2 = 0.90 J

Ek after = ½ × 0.20 × 1.5^2 + ½ × 0.30 × 1.56^2 = 0.225 + 0.364 = 0.59 J

0.59 J < 0.90 J, so kinetic energy is **not** conserved: the collision is **inelastic**. About 0.31 J went into heating and sound.

MARK-SCHEME WORDING

- Marks: "correct Ek for one object", "all Ek values correct", "**conclusion consistent with the calculated values**".
- "Elastic because total Ek before = total Ek after"; "not elastic because Ek before ≠ Ek after" (or "Ek before greater than Ek after").
- Near-equal answers are accepted as elastic: 9.62 × 10^{-3} J and 9.75 × 10^{-3} J "may be considered approximately equal".

Often asked as: the last part of nearly every 2-D collision question; MCQ "momentum conserved, kinetic energy not conserved".

COMMON MISTAKE

Leaving out the comparison sentence: it is a mark on its own.

Forgetting a body that was already moving before the collision, or adding only one of the energies after.

No unit (J) anywhere in the working.

Saying "energy is lost" in an inelastic collision. Total energy is conserved; **kinetic** energy is not.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Using the elastic condition (Ek before = Ek after) as an extra equation to find an unknown speed (WPH14 Jan 2026).
- Applying the test to particles, e.g. an alpha particle colliding with a proton, with energies around 10^{-12} J (WPH14 Jun 2023).

SPEC 86 Kinetic energy and momentum: Ek = p²/2m

WHAT THIS SPEC POINT ASKS

Derive and use Ek = p²/2m for a particle moving much slower than light.

Derivation

Start from Ek = ½mv² and p = mv, so v = p/m:

Kinetic energy of a non-relativistic particle

FORMULA SHEET

$$E_k = \frac{p^2}{2m}$$

E_k: kinetic energy / J p: momentum / kg m s^{-1} m: mass / kg

Deriving $E_k = p^2/2m$

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Start from the kinetic energy of a mass m moving at speed v:

$$E_k = \tfrac{1}{2}mv^2$$

2 Multiply the top and bottom by m. This changes nothing (it's multiplying by 1) but builds an mv on top:

$$E_k = \frac{m \cdot mv^2}{2m} = \frac{m^2v^2}{2m} = \frac{(mv)^2}{2m}$$

3 The bracket on top is the momentum, $p = mv$:

$$\boxed{E_k = \frac{p^2}{2m}}$$

Rearranged, $p = \sqrt{2mE_k}$. Use whichever direction the question needs.

- Ek in J, p in kg m s^{-1} (N s), m in kg.
- **Non-relativistic only.** If ½mv² gives a speed near or above c (3.00 × 10⁸ m s^{-1}), the formula is not valid.
- Use it whenever you know the momentum (from conservation, from r = p/BQ, or from λ = h/p) and want the energy, or the other way round.

Quick ratio results (for MCQs)

- Same momentum: Ek ∝ 1/m, so the **lighter** body has more kinetic energy (a recoiling nucleus has less Ek than the alpha particle).
- Same mass: Ek ∝ p². Halve the momentum and Ek falls to a **quarter**.

WORKED EXAMPLE

electrons with a chosen de Broglie wavelength

Electrons for a diffraction experiment need λ = 1.0 × 10^{-10} m. What accelerating p.d. is needed?

p = h/λ = 6.63 × 10^{-34} ÷ 1.0 × 10^{-10} = 6.63 × 10^{-24} kg m s^{-1}

Ek = p²/2m = (6.63 × 10^{-24})² ÷ (2 × 9.11 × 10^{-31}) = 2.41 × 10^{-17} J

eV = Ek → V = 2.41 × 10^{-17} ÷ 1.60 × 10^{-19} = **150 V** (2 s.f.)

WORKED EXAMPLE

momentum of a proton

A proton has Ek = 5.0 MeV. Ek = 5.0 × 10⁶ × 1.60 × 10^{-19} = 8.0 × 10^{-13} J.

p = √(2 × 1.67 × 10^{-27} × 8.0 × 10^{-13}) = **5.2 × 10^{-20} kg m s^{-1}**

WORKED EXAMPLE

A stationary nucleus emits an alpha particle of mass 6.64×10^{-27} kg with kinetic energy 7.7×10^{-13} J. The nucleus left behind has mass 3.69×10^{-25} kg. Calculate its recoil kinetic energy and the fraction of the

total energy carried by the alpha particle.

Show solution

Momentum of the alpha particle:

$$p = \sqrt{2mE_k} = \sqrt{2 \times 6.64 \times 10^{-27} \times 7.7 \times 10^{-13}} = 1.01 \times 10^{-19} \text{ kg m s}^{-1}$$

The total momentum was zero, so the nucleus recoils with the same size momentum in the opposite direction:

$$E_k = \frac{p^2}{2m} = \frac{(1.01 \times 10^{-19})^2}{2 \times 3.69 \times 10^{-25}} = 1.4 \times 10^{-14} \text{ J}$$

Fraction carried by the alpha particle:

$$\frac{7.7 \times 10^{-13}}{7.7 \times 10^{-13} + 1.4 \times 10^{-14}} = 0.98$$

Check with the mass ratio: $E_{\text{nucleus}}/E_\alpha = m_\alpha/m_{\text{nucleus}} = 0.018$, and $0.018 \times 7.7 \times 10^{-13} = 1.4 \times 10^{-14}$ J.

MARK-SCHEME WORDING

- "Use of Ek = p²/2m **Or** use of Ek = ½mv² and p = mv": both routes score.
- A separate mark for converting eV (or MeV) to J.
- "Show each step and say which equation" in a de Broglie show-that.

Often asked as: combined with r = p/BQ to find B, combined with λ = h/p to find V, mass of an ion from p and Ek, Ek after a collision from momenta, MCQ on how Ek changes when p changes.

COMMON MISTAKE

Using the electron mass for a proton (or the other way round).

Dropping the "M" in MeV, or dividing by 1.6 × 10⁻¹⁹ when you should multiply.

The formula p = √(2mEk) is not in every formula list, so learn the derivation.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **de Broglie wavelength** λ = h/p (from Unit 2) is combined with Ek = p²/2m and Ek = eV almost every year (WPH14 Jan 2024, Jan 2025).
- Showing that ½mv² gives v > c for a 23 MeV electron, so a relativistic treatment is needed (WPH14 Jun 2021).

Quick checks

QUICK CHECK

A 0.50 kg ball hits a wall at 4.0 m s^{-1} and bounces straight back at 4.0 m s^{-1}. What is the magnitude of its change in momentum?

- 0 N s
- 2.0 N s
- 4.0 N s
- 8.0 N s

The speed is unchanged but the velocity reverses: $\Delta p = 0.50 \times (-4.0 - 4.0) = -4.0$ N s. Zero is the trap: momentum is a vector, so a change of direction is a change of momentum.

QUICK CHECK

Why does an airbag reduce the force on a driver's head in a crash?

- It reduces the head's change in momentum.
- It increases the time taken for the head to stop, so a smaller force gives the same impulse.
- It absorbs the head's momentum so momentum is not conserved.
- It reduces the impulse on the head.

The head goes from moving to stopped either way, so Δp (and the impulse) is the same. $F = \Delta p / \Delta t$, so a longer Δt means a smaller average force.

QUICK CHECK

Puck A moves along the x-axis and hits stationary puck B. Afterwards B moves off at 30° below the x-axis. Which statement about A after the collision must be true?

- A moves along the x-axis.
- A has a component of velocity above the x-axis (or A is at rest).
- A moves at 30° below the x-axis.
- A must stop.

Before the collision the total y-momentum is zero, so afterwards A's y-momentum must cancel B's downward component. The only way A could avoid this is by having no momentum at all.

QUICK CHECK

Particles X and Y have the same momentum. X has four times the mass of Y. What is the ratio (kinetic energy of X) : (kinetic energy of Y)?

- 1 : 4
- 4 : 1
- 1 : 2
- 1 : 16

$E_k = p^2/2m$. With the same p, kinetic energy is inversely proportional to mass, so the heavier particle has a quarter of the energy.

QUICK CHECK

Two trolleys collide and stick together. No external forces act. Which statement is correct?

- Total kinetic energy and total momentum are both conserved.
- Total kinetic energy is conserved but total momentum is not.
- Total momentum is conserved but total kinetic energy is not.
- Neither is conserved, but total energy is.

With no external force, momentum is always conserved. Objects sticking together is always inelastic, so some kinetic energy is transferred to thermal energy and sound. Total energy is conserved too, which is why the last option is a trap: momentum is conserved as well.

QUICK CHECK

Which of these is not a correct unit for impulse?

- N s
- kg m s^{-1}
- N s^{-1}
- J s m^{-1}

Impulse is force × time, so N s, which equals kg m s^{-2} × s = kg m s^{-1}. Since 1 J = 1 N m, J s m^{-1} is also N s. N s^{-1} would be a rate of change of force.